

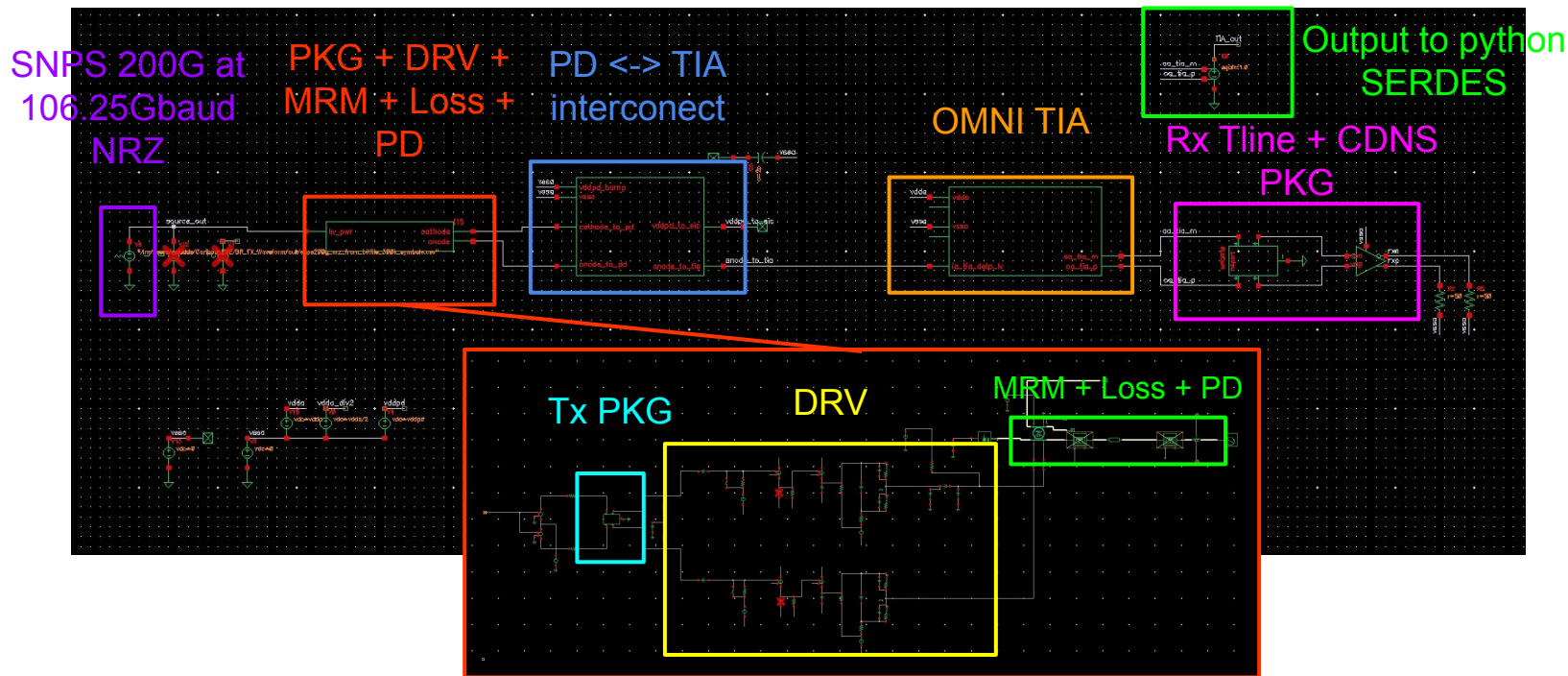
NVDA EOE Work

Alex Sludds



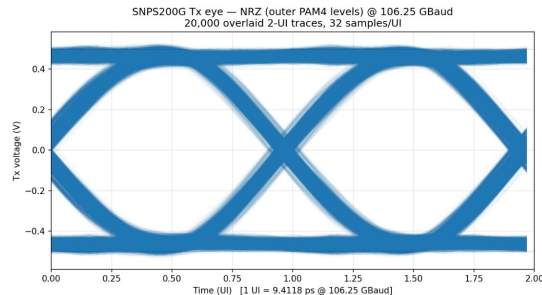
Testbench

- Semi-Behavioral DRV used due to errors in Omni DR driver at time of simulation
- CDNS PKG used on Rx since it gives waveform after SERDES T-coil + AFE (more representative of what the python model seems to consume)

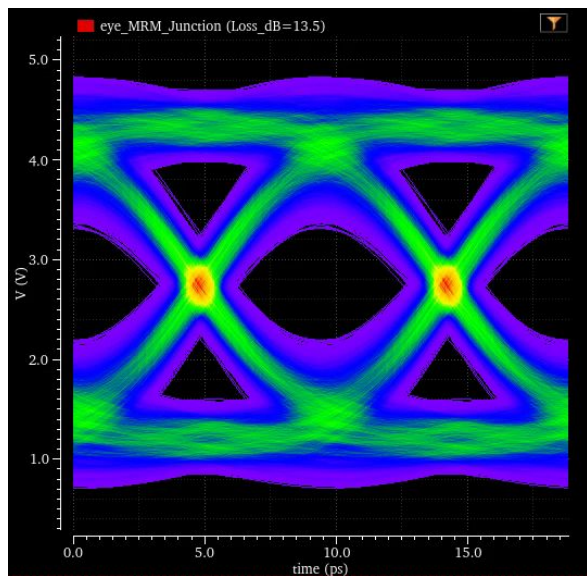


Testbench Results

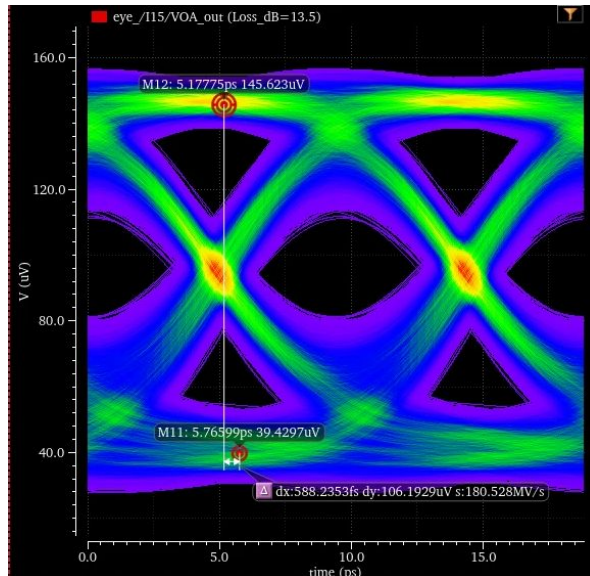
- No Tx Eq, 1V_{pp} SERDES swing (quick python plot shown right)



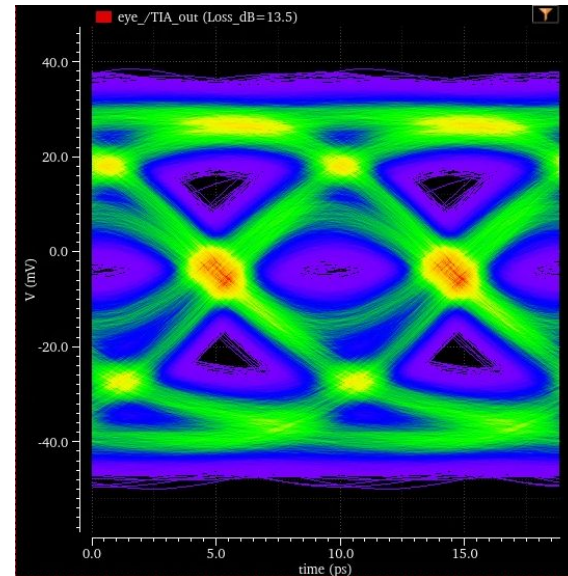
MRM Junction Eye



PD Input Eye @ 106uW OMA

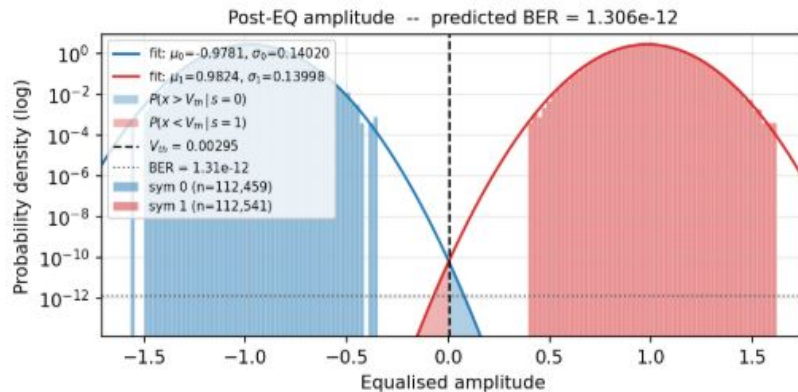
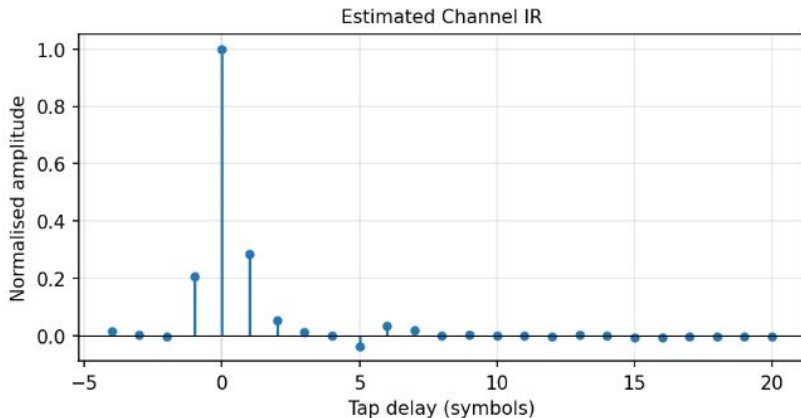
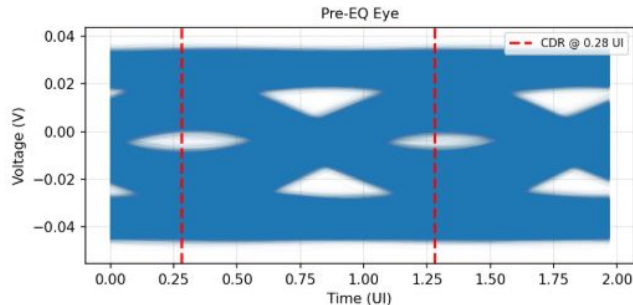


TIA Output Eye @ 106uW OMA



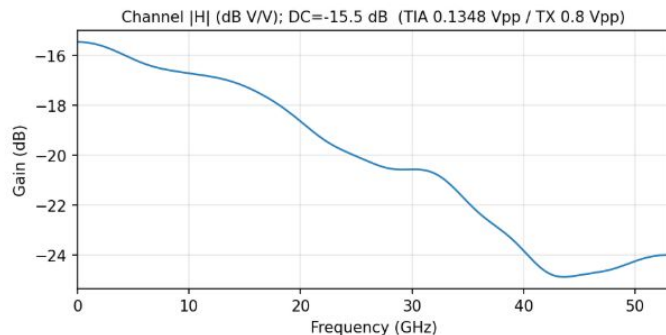
Python SERDES Results

- Results are for 13.5dB loss case -> 106uW OMA
- Assume Rx SERDES has:
 - 2 pre-cursor FFE taps, 2 post-cursor FFE taps
 - 1 DFE tap
 - No CTLE
 - Muller-Muller CDR
- Final FFE taps: [-0.0709, -0.1019, 2.0931, -0.449, 0.0878]
- Final DFE taps: [0.2217]

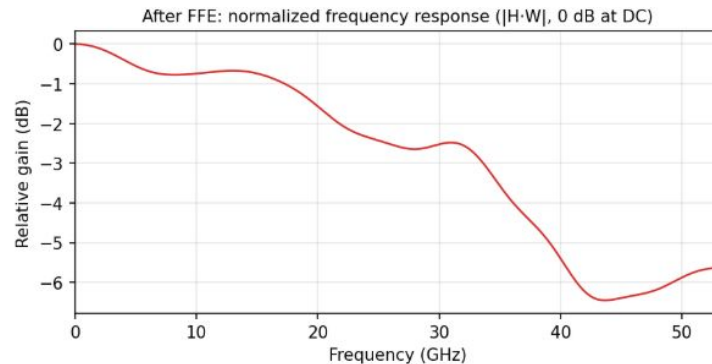


Python SERDES Results: Cont

Pre-Eq Estimated Channel
Frequency Response



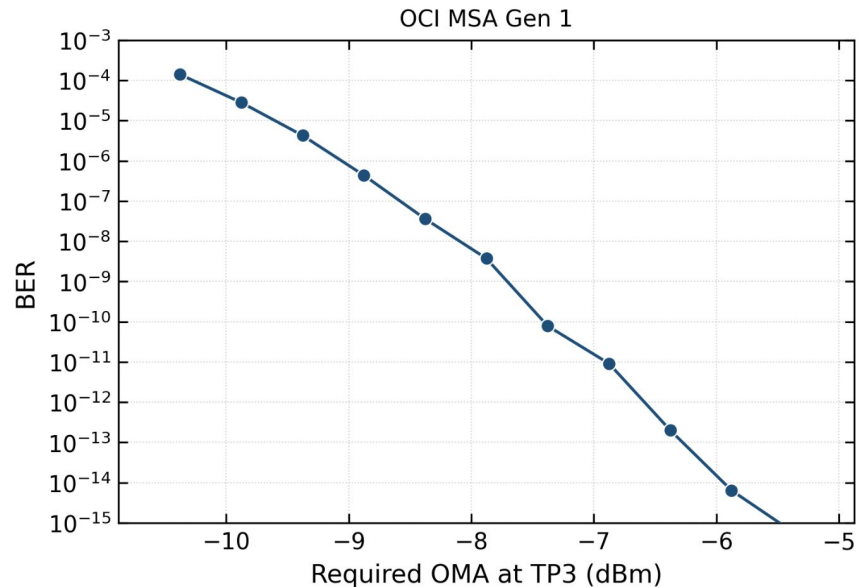
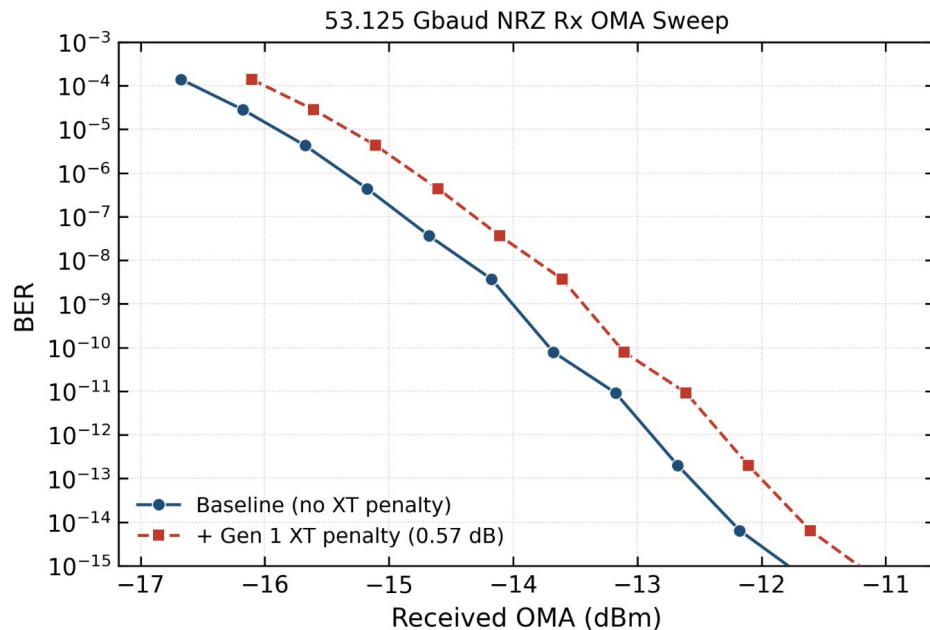
Post-FFE Estimated Channel
Frequency Response



Note: does not include DFE (can't add a non-linear equalizer to a frequency response plot)

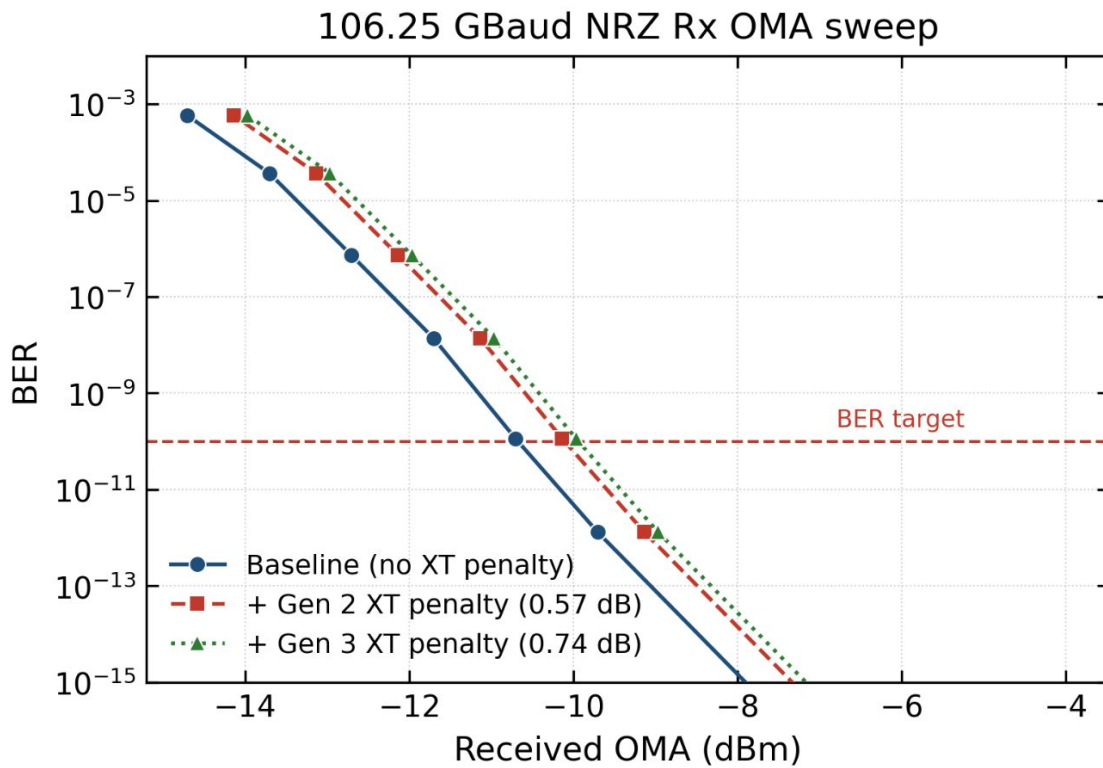
Channel bandwidth extended from ~23GHz to ~33GHz via FFE, loss at nyquist also becomes more tame (10dB -> ~5dB)

Rx Sensitivity: OCI Gen 1



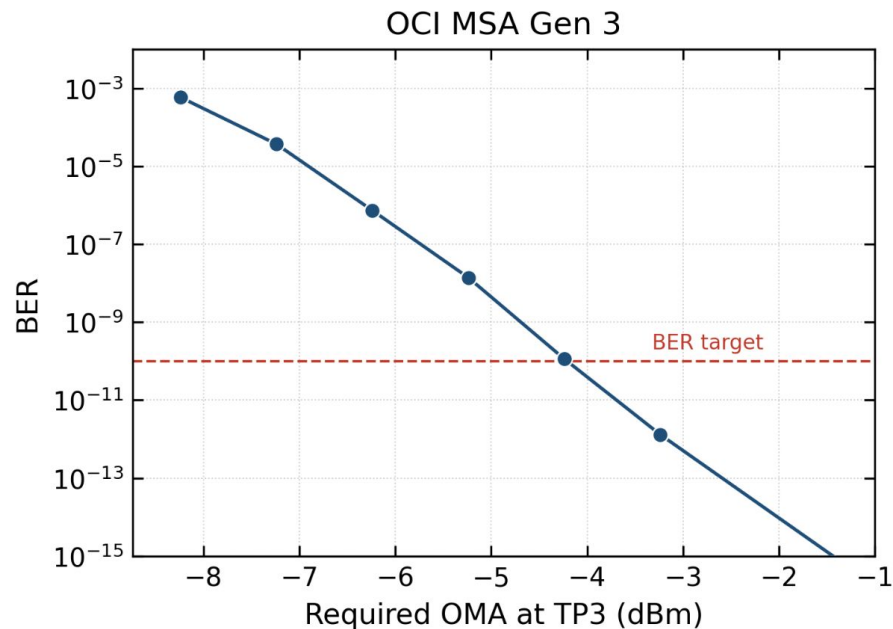
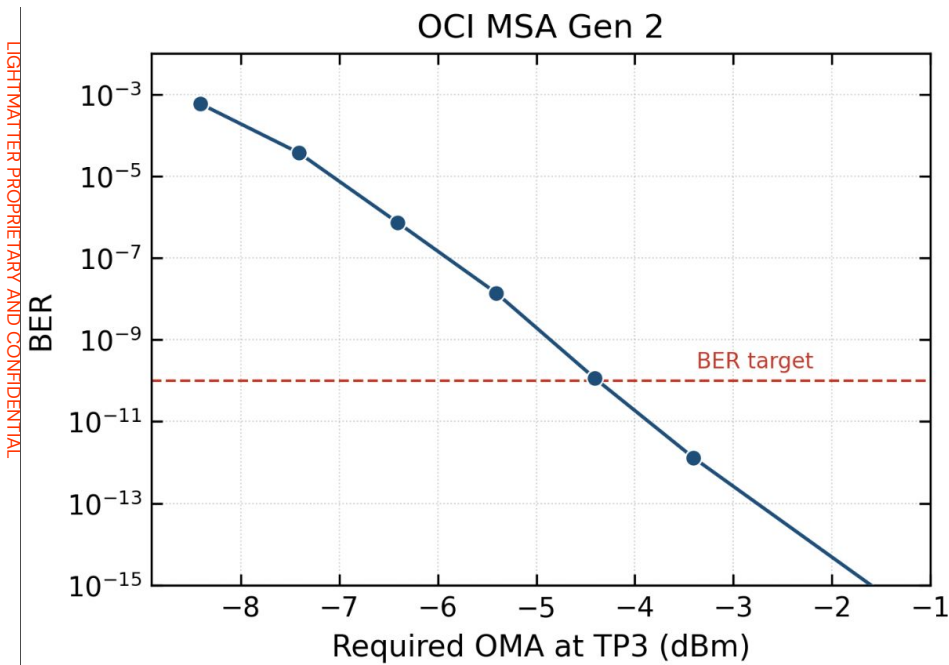
Rx Sensitivity: OCI Gen2 + Gen 3

- Measured over 300k symbols of PRBS31
- Internal image, do not share
- Required OMA without XT = -10.61 dBm
- Required OMA with Gen 2 XT = -10.0 dBm
- Required OMA with Gen 3 XT = -9.87 dBm



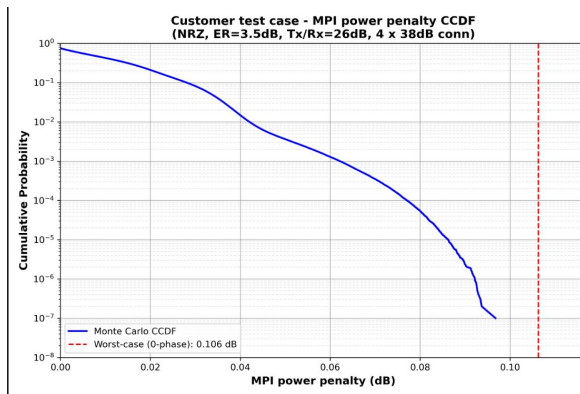
Rx Sensitivity

- Spectral and coherent crosstalk power penalty added



MPI Power Penalty

- Two estimation methods used, most conservative taken:
- [Bhatt](#) which gives a absolute worst case value for a deployment (assumes everything goes wrong)
 - Power penalty is 0.276dB
 - I assume 26dB Tx and Rx reflectance, 38dB return loss from 4 connectors. 3.5dB modulator ER. NRZ.
- [King](#) which uses a statistical model to estimate power penalty. It's generally more relaxed than Bhatt since it's statistically unlikely to get everything wrong at the same time
 - Power penalty at 10^{-6} probability (if I build 1 million fiber links, I expect 1 will have this penalty) is ~0.09dB. Worst case 0.106dB for king (set all phases to zero).

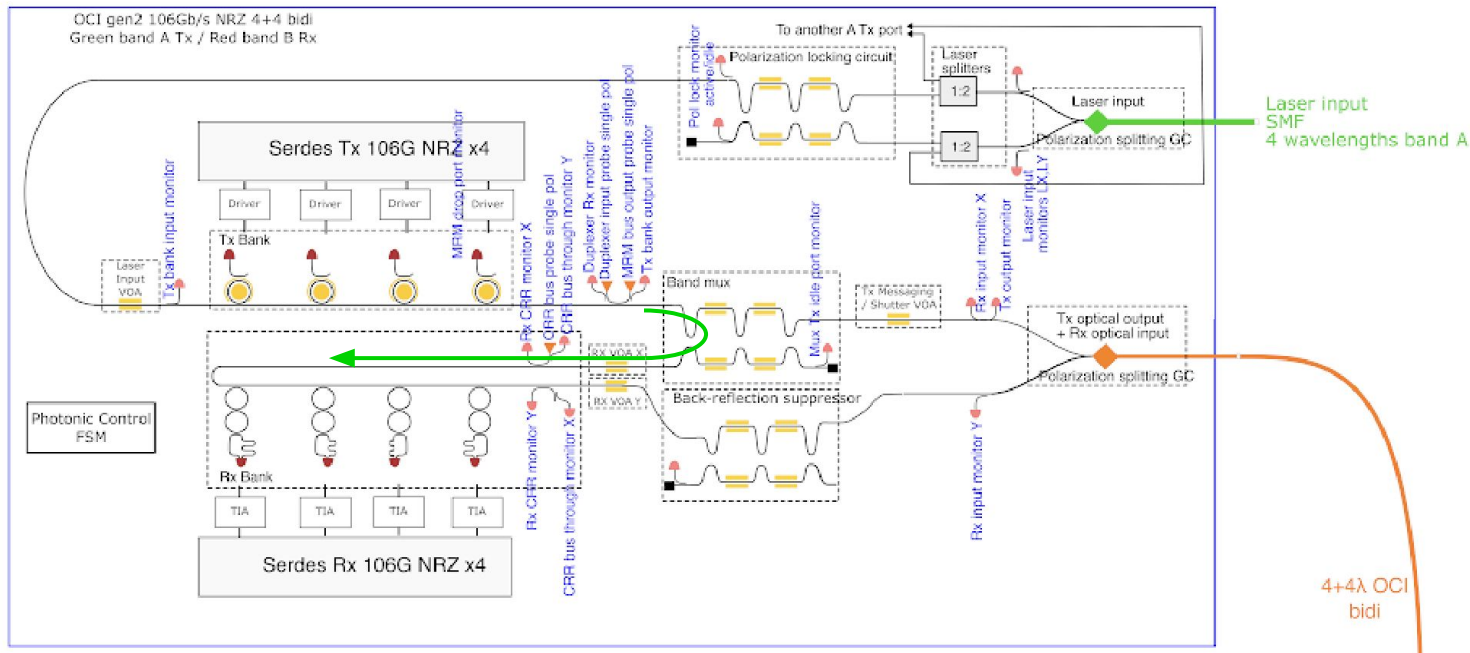


Spectral Crosstalk

- Let's identify crosstalk terms we think will be problematic and quantify which one dominates
- [The math for these cases is done here](#)

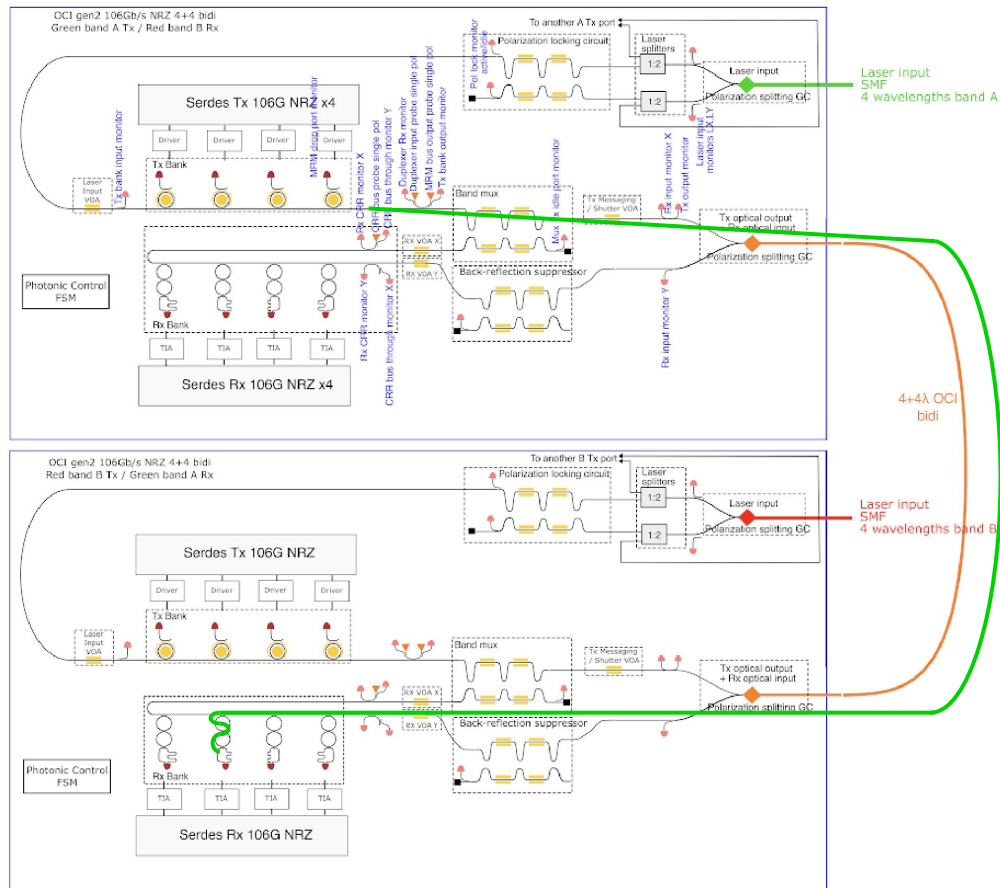
Spectral Crosstalk: Term 1

- Reflection off of the interleaver input
- Light hitting PD is MRM launch + interleaver reflection + CRR stop band rejection
 - Potential for this light to end up in CRR passband depending on CRR design



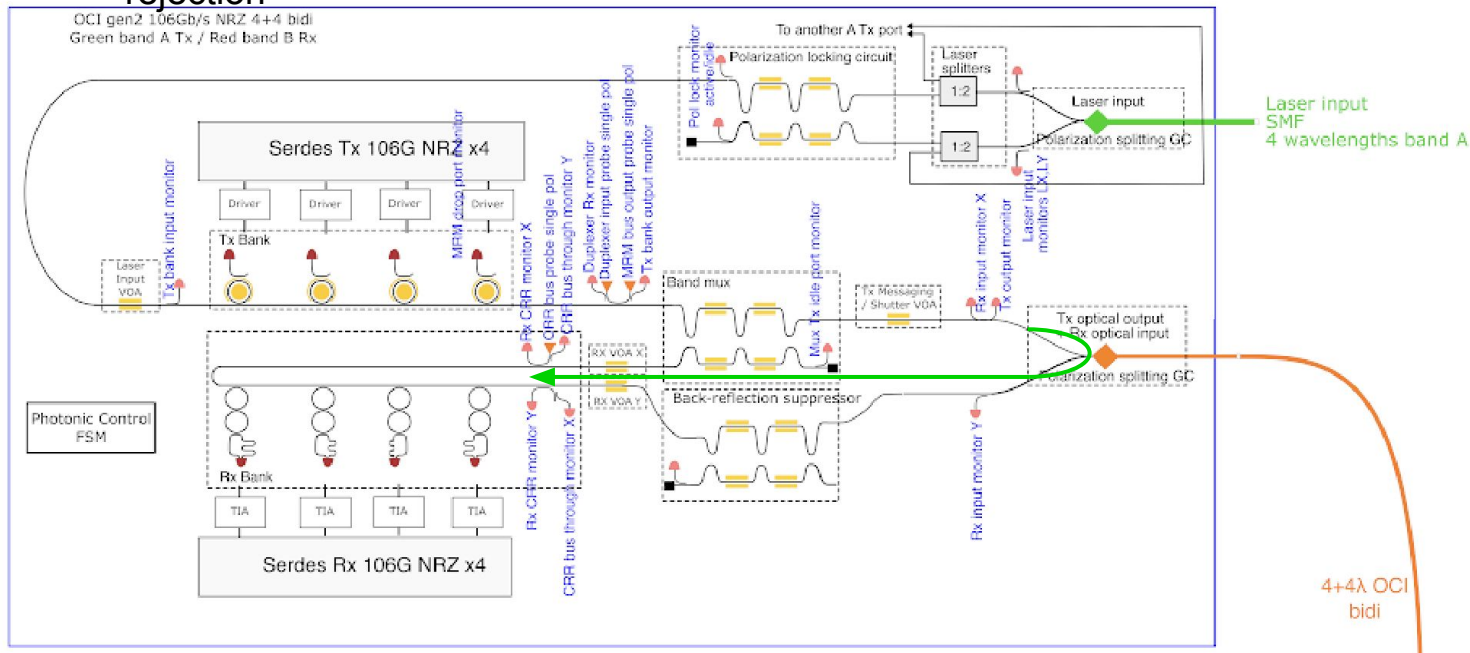
Spectral Crosstalk: Term 2

- Standard spectral crosstalk
- Light hitting PD is PD Rx OMA + CRR stop band rejection



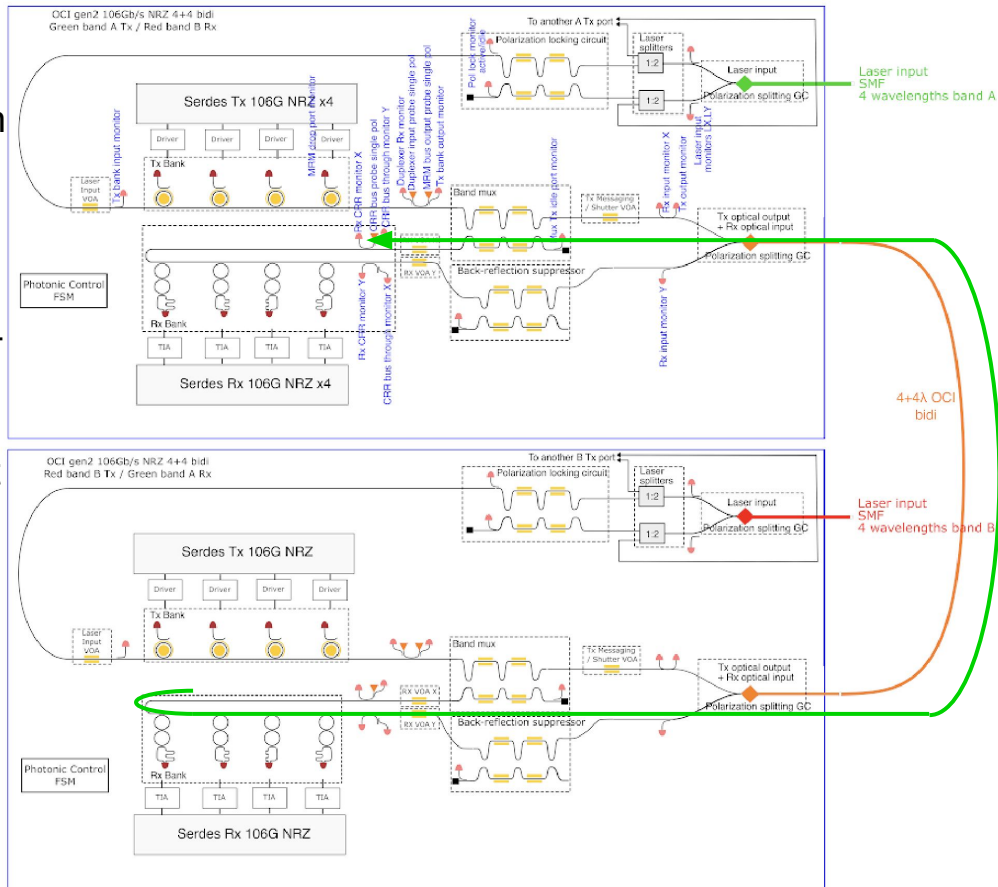
Spectral Crosstalk: Term 3

- Tx launch light reflects off something after the grating (or the grating itself), ends up in the linear polarization that sends all light down the delay comp path
 - Crosstalk is Tx launch OMA + ORL + back reflection suppressor rejection + CRR rejection



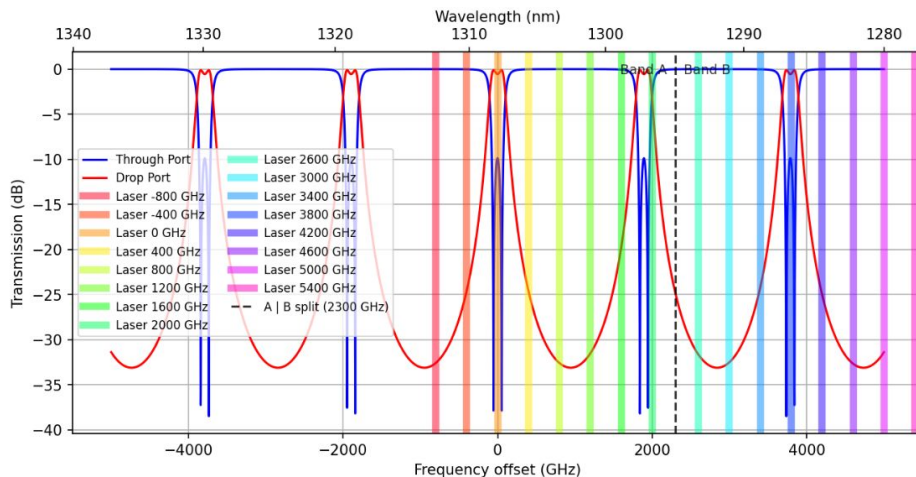
Spectral Crosstalk: Term 4

- Light comes to the CRR, not all of it is perfectly dropped ($\sim 10\text{dB}$ leakage based on allowable ripple)
- Some of that makes its way back to Rx bank on the other chip
- Crosstalk amount is equal to:
 - Rx OMA + CRR leakage + path loss + interleaver rejection + CRR loss (light could be in passband)
- Note that Rx side duplexer passes light, but Tx side does not



Spectral Crosstalk: CRR Design

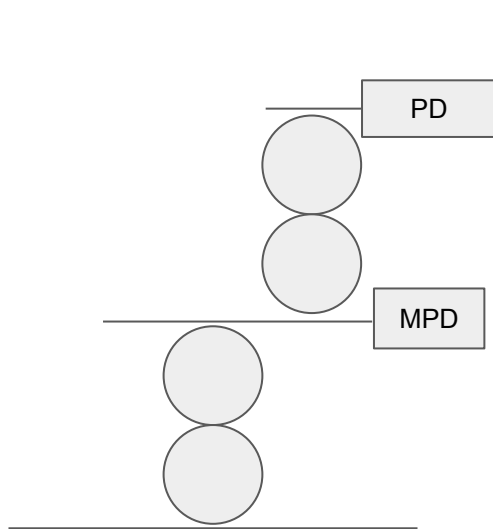
- The plot below shows the guess at the OCI MSA Gen 3 wavelength bands (thank you Mike)
 - We don't have a big guard band between wavelength sets, can't design for a CRR resonance to land in the guard band for all wavelengths
 - Can't make a CRR with 6000GHz FSR (probably 1.5-2um radius?)
1. Double CRR
 - a. More area/control. A bit more loss. "Keep it simple stupid"
 2. Vernier CRR
 - a. Very very large FSR, crosstalk term from spurious peaks must be kept under control
 - b. Reliability issues if one of the rings needs to tune an extra full FSR to lock to a target



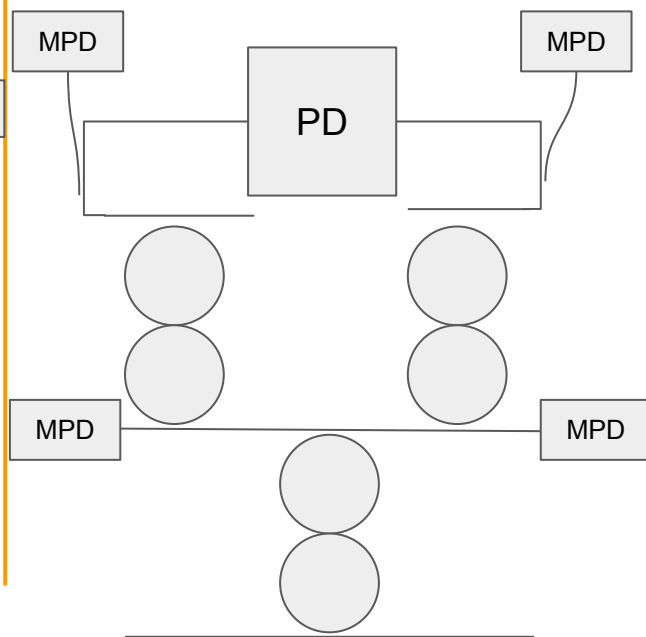
Spectral Crosstalk: Double CRR

- A double CRR is the idea of stacking two CRRs one after the other, mismatching their FSR
- You run into issues of polarization diversity (our CRRs expect signals from both left and right bus ports)

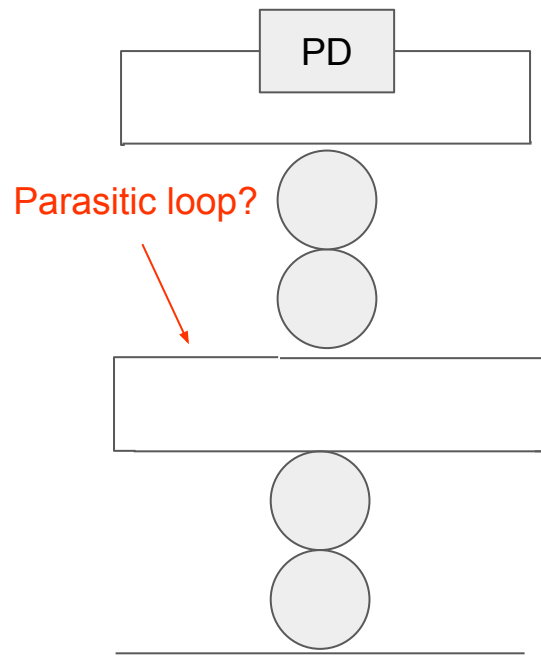
Core idea
Has no pol diversity



Nieve idea to support pol diversity
Has issues with CRRs going dark



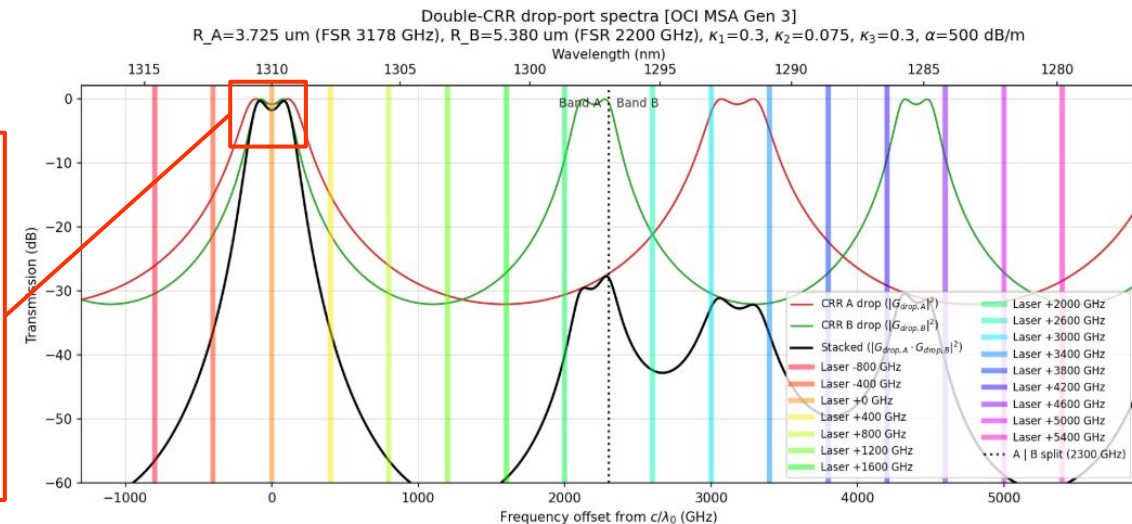
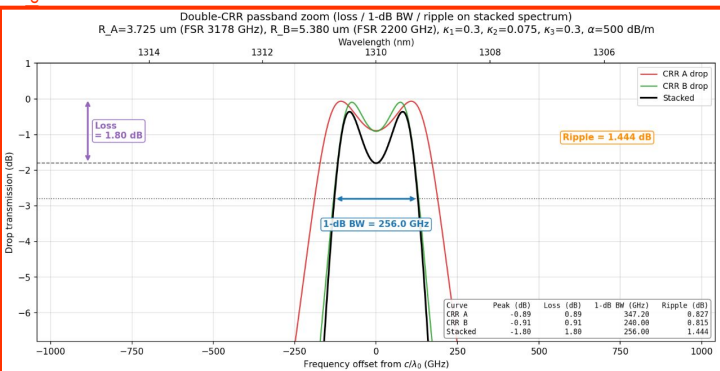
Reuse the second CRR



Spectral Crosstalk: Double CRR

- Here is a toy design for this system (I'm sure someone can do better than me)
- First CRR is designed to have FSR > band A wavelengths ($8 \times 400 = 3200$)
 - This is so that we don't drop two desired wavelengths from the bus
- Second CRR can have smaller FSR to do filtering

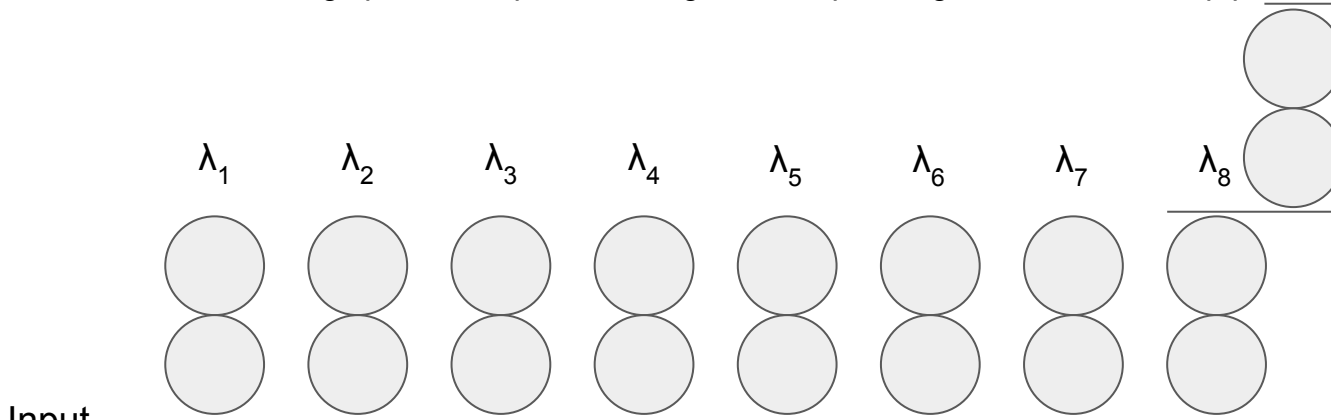
LIGHTMATTER PROP



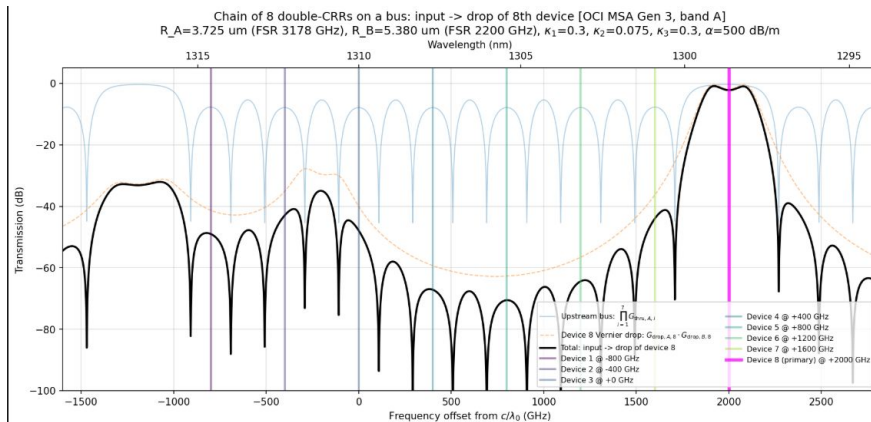
Spectral Crosstalk: Double CRR

- Double checking spectral response through a bus up through double CRR drop port

Output



Input



Spectral Crosstalk: Vernier CRR

- Suppose I don't want to do double CRR (due to extra power, area, loss etc)
- Vernier CRR has FSR given by $|FSR_1 * FSR_2| / |FSR_1 - FSR_2|$
- Reasonable ring FSR is ~1000GHz to 2000GHz
- We know Vernier CRR needs FSR > 6000GHz
 - Need a "Gain" ($FSR_2 / |FSR_1 - FSR_2|$) of ~6 -> $FSR_2 = 7/6 * FSR_1$
- Suppose $FSR_1 = 1200\text{GHz}$ and $FSR_2 = 1400\text{GHz}$ (nice round numbers)